

Experts' views

"The Indian strategic electronics segment today, due to various global and domestic factors, is faced with a unique situation of quantum jump in opportunities in manufacturing. There is an urgent need in the short term to provide for easy access to finance to support cash flow and fund these new opportunities. The government should immediately look at coming out with a comprehensive financial package and putting a mechanism to ensure that access to finance is never an issue for developing the electronics manufacturing ecosystem."

—**Col. Sharath Bhat (Retd), senior vice president, Kaynes Technology**

"DRDO labs have been doing R&D for defence for a long time. Over a period, they have developed many crucial technologies that are useful not only for defence but also find application in the industry. But DRDO labs are not marketing these technologies to the industry aggressively. We need to exploit these technologies to the maximum possible extent so that the country can avail the best value for the efforts put in by DRDO labs. I feel we should have an independent central agency for this purpose, whose agenda should be to transfer the technology developed by these labs to industry expeditiously on simple terms and conditions purely on commercial basis, as is done by C-DOT."

—**Subhash Goyal, managing director, Digital Circuits Pvt Ltd**

"Clarity on the financial model for transfer of technology has to be clearly defined for technology transfer from DRDO labs to the private sector. MSMEs hesitate a lot to pay for transfer of technology without clear production assurance. A small upfront fee (not hurting MSMEs) with a licensing model would be ideal. Also, preference should be given to the vendor/MSME who has developed the initial product, while working closely with transfer of technology."

"Moreover, a balance between innovation and L1 tendering has to be obtained to ensure ease of doing business for MSMEs in the strategic electronics segment. R&D and innovation cannot be always successful in L1 process. However, an alternative approach is not easy in a country of our size."

—**Rajeev Ramachandra, chief technology officer, Mistral Solutions**

"It is an acknowledged fact that India's space programme has been a notable success. However, exploitation of this programme has largely remained bound to domestic utilisation, except for commercial launches of a few foreign satellites. Recently, ISRO has engaged the private industry to help augment its satellite-building capacity. With infrastructure, skilled manpower and R&D facilities created over the years, both within ISRO and the expanding private sector, there is a tremendous opportunity and need to scale up commercial exploitation by exporting space technologies and services to the huge global market. This needs to be implemented at the earliest, lest we miss the bus with increasing competition from other countries. Setting up of New Space India Pvt Ltd under Department of Space, in addition to the existing Antrix Corp., will hopefully help exploit this potential."

—**Lt Gen. (Dr) Ajay Chande (Retd), honorary member and senior advisor, ELCINA and governing council member, IETE**

to operate within the defence industry, specifically in the areas of electronics system design, original design manufacturing (ODM), etc.

The government has already launched some programmes and schemes to facilitate R&D activities in this field. Technology Development Fund (TDF) and recently-opened Army Design Bureau are worth mentioning here. TDF is bound to enable and enhance R&D activities through the expansion of industry-academia partnership, and will be a big push for Make in India programme in the defence sector. And, Army Design Bureau can work in sync with Defence Research and Development Organisation (DRDO) and the private industry to achieve modernisation and indigenisation. This, in turn, will enhance opportunities for public-private partnership.

DRDO labs are practically the sole R&D agencies for design of defence weapons and equipment. Products designed by them are invariably transferred to defence PSUs for manufacture. The private sector gets very little transfer of technology from DRDO labs. This needs to change. **The private sector, including MSMEs, needs to be given a fair share of technologies for product development by DRDO labs. They could be involved in the design stage, which would ensure seamless transfer of technology.** Where the private sector lacks infrastructure and test facilities, these should be provided by the DRDO/DPSUs. This will ultimately help in the creation of a viable defence manufacturing capability in the private sector and promote self-reliance in the defence sector.

It is also recommended that pro-

totype development by the private sector be funded, preferably with funds from the budget of the user defence service, which will create a sense of ownership. The recipient service would then closely monitor the development process and may even be able to suggest modifications, if required.

Policy boost

In the last few years, the strategic electronics sector got enough policy boost through the amendment of DPP, which included the removal of 49 per cent foreign direct investment (FDI) limit in the defence sector. Key provisions of DPP 2016 are:

- Inclusion of Buy (Indian-IDDM) procurement category
- Higher but flexible rule for indigenous content requirement
- Revamped procedures for Make projects
- Institutionalisation of RFI process
- Introduction of L1-T1 methodology for selecting final bidder
- Increased threshold limit for offsets

Removal of 49 per cent FDI limit in the defence sector has opened up avenues for significant investments and has given the much required impetus for many SMEs to have access to finance and technology. The liberalised FDI regime permits up to a hundred per cent (up to 49 per cent through the automatic route and beyond that through Foreign Investment Promotion Board, or FIPB, route) foreign equity in the defence sector, as India looks ahead to increase indigenous manufacturing in defence under Make in India programme.

However, FDI policy cannot become the only instrument to attract investment. It will be successful only if foreign OEMs can derive the necessary confidence in the system from an operational point of view. Moreover, foreign OEMs are less likely to create intellectual property (IP) in India. Therefore Design in India concept will not be promoted through joint ventures with higher FDI.

The Indian Ministry of Defence (MoD) has issued a new policy on